

Sulfur-Doped IrO₂ Enable Pathway Switch to Lattice Oxygen Mechanism with Enhanced Stability for Low Iridium PEM Water Electrolysis

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Achieving high activity and stability while minimizing Ir usage poses a significant challenge in the industrialization of proton exchange membrane water electrolysis (PEMWE). Herein we report a sulfur-doping strategy that enables the OER pathway on IrO₂ nanoparticles (IrO₂/S) to switch from conventional adsorption evolution mechanism (AEM) to lattice oxygen mechanism (LOM) while maintaining Ir–O bond stability, thus achieving a significant enhancement in both intrinsic activity and durability. Advanced spectroscopies and theoretical calculations reveal that the Ir–S coordination motif within the lattice increases the electron density of the Ir center and enhances Ir–O covalency, thus triggering the LOM pathway. Importantly, the lattice distortion and unsaturated Ir–O coordination within the IrO₂/S generate the oxygen nonbonding state that acts as an electron sacrificial agent to preserve Ir–O bonds upon the LOM-dominated OER process. As a result, PEMWE integrated with such IrO₂/S electrocatalyst delivers a low cell voltage (1.769 V at 2.0 A cm^{−2}) and long-term stability (16.6 μV h^{−1} over 1000 h@1.0 A cm^{−2}) while dramatically reducing Ir usage from 1.0 to 0.3 mg cm^{−2}. This work establishes S doping as a viable strategy to trigger LOM and stabilize lattice oxygen redox in Ir-based catalysts, opening a new avenue for low-Ir PEMWEs.

1. Introduction

The development of highly active and durable Ir-based oxidized catalysts is an essential way to achieve cost-effective industrial applications of proton exchange membrane water electrolysis (PEMWE).^[1] Previous studies have demonstrated that the oxygen evolution reaction (OER) process on the typical Ir/IrO₂ catalysts commonly follows an adsorption evolution mechanism (AEM), that is restricted to the scaling rule, leading to the limited intrinsic activity.^[2] Lattice oxygen mechanism (LOM), another classic reaction pathway for the OER, involves the direct participation of lattice oxygen atoms, which can break the linear scaling relationship and exhibit extremely high intrinsic OER activity. Thus, switching the OER pathway to LOM appears to be a promising strategy for dramatically enhancing the OER activity of the IrO₂ catalyst. Based on the molecular orbital theory, pushing the oxygen bands toward the Fermi level can trigger the LOM mechanism, when oxygen redox chemistry works instead of metal

redox chemistry in AEM.^[3] Transition metal doping,^[4] which can effectively modulate the energy level of O atom within the IrO₂, has been reported as a feasible approach to realize the occurrence of LOM on IrO₂ catalyst. For example, the trivalent metals (Gd, Nd, Pr) doped IrO_{2−δ} can intensify the Ir–O covalency to activate the lattice oxygen by narrowing the energy gap between Ir 5d and O 2p band center, which is beneficial for triggering the LOM pathway.^[5]

Though the above mechanism switch strategy has achieved significant progress in terms of OER activity, it sacrifices structural stability. As known, the main challenge regarding the LOM is its inherently poor stability. When the lattice oxygen directly participates in the OER cycle, the electron loss from the M–O bond would lead to a decreased M–O bond order, and hence to the declined structural stability.^[5b] Therefore, inhibiting electron loss from Ir–O holds great promise for achieving a robust structure while following the LOM pathway on IrO₂ catalysts. Recent studies have demonstrated that constructing oxygen nonbonding (O_{NB}) states represents a highly effective strategy to stabilize the LOM pathway.^[3b,6] For instance, nitrogen doping in cobalt

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DOI: 10.1002/adma.202507560

oxide can create O_{NB} states, stabilizing the Co–O bond during the LOM-dominated OER process.^[7] Similarly, Zn^{2+} doping in CoOOH also induces the formation of O_{NB} states, enhancing the durability of the catalyst.^[6a] However these materials are primarily concentrated on complex perovskite and spinel structure systems, where intricate coordination environments, flexible spatial configurations, and multicomponent synergies facilitate the formation of O_{NB} . By contrast, engineering O_{NB} states in single-component oxide—hindered by rigid lattices and limited multicomponent synergies—remains underexplored. Especially for IrO_2 , concurrently inducing O_{NB} and switching to LOM within its simple framework has been rarely explored.

To bridge this gap, we report a dual-functional strategy of sulfur lattice doping, enabling a switch of the OER pathway from AEM to LOM while stabilizing IrO_2 catalysts (Figure 1a). Structural characterizations and theoretical calculations show that S doping induces lattice expansion and unsaturated Ir coordination, restructuring the energy landscape of oxygen to form O_{NB} , and enabling a cooperative enhancement of both activity and stability. In situ Raman spectroscopy and electrochemical impedance spectroscopy (EIS) demonstrate that S doping accelerates active site oxidation and electron transfer, thereby enhancing OER kinetics. In situ electron paramagnetic resonance (EPR) and operando differential electrochemical mass spectrometry (DEMS) directly demonstrate the occurrence of the LOM pathway on the IrO_2/S catalyst. As a consequence, the catalyst achieves a low cell voltage (1.769 V at 2.0 A cm⁻², Nafion 115) with a greatly reduced Ir loading (0.3 mg cm⁻²) and good stability (16.6 μ V h⁻¹ decay over 1000 h) in PEMWE, suggesting S doping as an effective approach to unlock LOM in IrO_2 for practical PEMWE applications.

2. Results and Discussion

2.1. Synthesis and Structural Characterization

We designed S as a dopant because it belongs to the same main group element as O, which might enable substitutional doping (via similar valence electron configuration) to form stable Ir–S bonds, optimize electronic structure through favorable 3p–5d orbital hybridization, and balance lattice compatibility. The S-doped IrO_2 nanoparticles were synthesized via a modified Adams fusion method, where thiourea served as both S source and complexing agent.^[8] The formation of an Ir–S complex during the synthesis was confirmed by a color change from dark brown ($IrCl_3$) to opaque yellow upon thiourea addition at 80 °C (Figure S1, Supporting Information). This discrepancy is attributed to the molecular architecture of thiourea, where the S atom—equipped with lone-pair electrons and a C=S double bond—functions as a soft-base ligand, aligning with the hard-soft acid-base (HSAB) theory to form stable covalent Ir–S bonds with Ir^{3+} (a soft acid). UV–vis diffusion reflectance spectroscopy (DRS) further validated the complexation, showing distinct absorption features for the Ir–S complex compared to pure $IrCl_3$ and thiourea (Figure S2, Supporting Information).^[8a] A series of IrO_2/S samples were obtained at different heat-treatment temperatures (denoted as $IrO_2/S-x$, $x = 300, 350, 400, 450$ and 500 °C, respectively). The atom percentage of S element is quantified by energy-dispersive X-ray spectroscopy (EDS) (Figures S3–S7 and Table S1, Support-

ing Information) and X-ray photoelectron spectroscopy (XPS) (Table S2, Supporting Information).

The XRD patterns (Figure 1b) reveal a clear evolution of crystallinity with annealing temperature. At 300 °C, the amorphous phase dominates due to incomplete oxidation of Ir–S complexes. At this stage, the calcined sample displays a yellow color (Figure S8, Supporting Information). When the annealing temperature increases from 350 to 450 °C, the IrO_2/S catalysts show a broad diffraction peak at $\approx 34.7^\circ$ in their XRD patterns, corresponding to the (101) crystal plane of IrO_2 (JCPDS no. 15–0870). Notably, upon magnifying the peaks from 25 to 45° in Figure 1b, it is observed that after S doping, the 2θ peaks in the (101) direction shift $\approx 0.5^\circ$ toward a lower angle. This shift occurs because O^{2-} ions (1.30 Å) are gradually replaced by large S^{2-} ions (1.84 Å), resulting in a lattice expansion of IrO_2 (Figure 1c).^[9] When the heat-treatment temperature reaches 500 °C, S is completely oxidized, and the (101) diffraction peak returns to 34.7°, confirming the reversible nature of the lattice expansion. Typically, this lattice expansion was also corroborated by Raman spectroscopy (Figure 1d; Figure S9, Supporting Information), where the E_g and B_{2g}/A_{1g} bands of $IrO_2/S-450$ shift to 538/703 cm⁻¹ from 548/716 cm⁻¹ of IrO_2 , reflecting the altered Ir–O bond vibrations.^[10] The high-resolution transmission electron microscopy (HRTEM) for a typical sample of the $IrO_2/S-450$ (Figure S10a, Supporting Information) indicates the formation of ultra-small nanoparticles with an average diameter of only $ca. 1.15 \pm 0.3$ nm. The d-spacing for the adjacent lattice fringe of the STEM image (Figure S10b, Supporting Information) of $IrO_2/S-450$ is $ca. 0.267$ nm, slightly larger than that of IrO_2 (0.258 nm),^[11] further demonstrating that S doping causes the lattice expansion of IrO_2 . Additionally, EDS mapping of $IrO_2/S-450$ shows that the Ir, O, and S elements are homogeneously distributed, indicating the successful synthesis of S-doped IrO_2 (Figure S10c, Supporting Information).

The X-ray absorption near edge structure (XANES) and extended X-ray absorption fine structure (EXAFS) at the Ir L_{III} edge were carried out to explore the electronic structure and the local environment of Ir. Figure 1e presents the Ir L_{III} edge XANES spectra of $IrO_2/S-x$, interestingly, the variation in the oxidation state of Ir with respect to the reaction temperature follows an anomalous trend. In particular, as the annealing temperature increases from 350 to 450 °C, the oxidation state of Ir decreases, likely due to the electron donation effect of S (lower electronegativity than O).^[9a,12] However, when the temperature reaches 500 °C, the disappearance of S causes the oxidation state of Ir increases and matches well with IrO_2 . This anomalous phenomenon in the oxidation state of Ir implies that the presence of S has a significant influence on the electronic configuration of IrO_2 . FT-EXAFS spectra at the Ir L_{III} edge (Figure 1f,g) and its fitting results (Figures S11 and S12, Supporting Information) uncover the atomic-scale structural changes induced by S doping. For pure IrO_2 , the dominant peak at 1.45 Å corresponds to Ir–O bonds with a coordination number (CN) of 5.9 and bond length of 2.00 Å, consistent with the rutile structure (Table S5, Supporting Information).^[13] In contrast, $IrO_2/S-450$ exhibits a reduced Ir–O coordination (CN = 4.5) and a slightly shortened bond length (1.99 Å), indicative of a significant structural distortion. This reduction might arise from two synergistic effects: First, S substitution introduces lattice strain, creating unsaturated Ir sites.^[14] Second, the electron donation effect of S weakens the Ir–O bonds

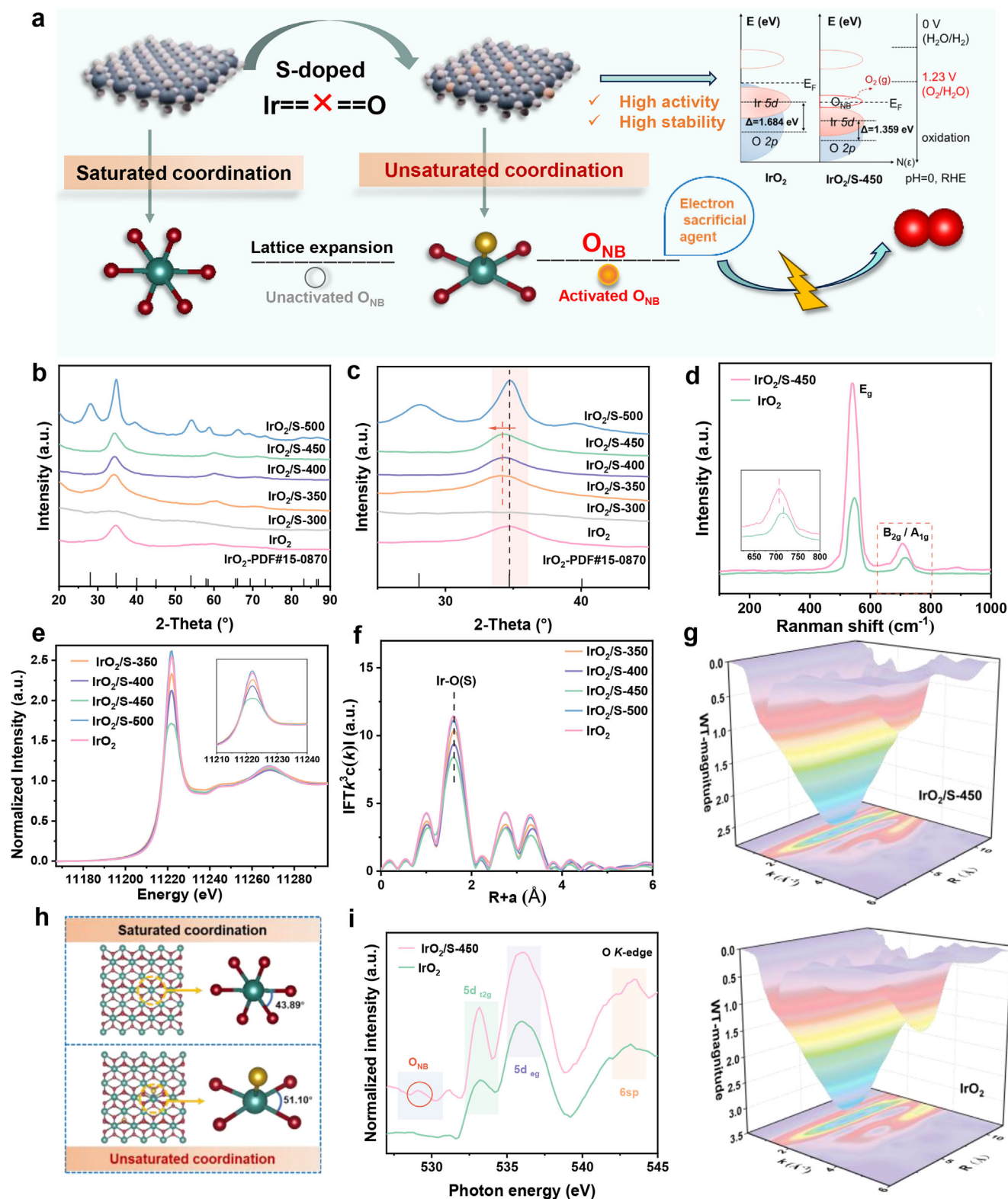


Figure 1. Morphology and structural characterization. a) Schematic illustration of the catalyst design strategy, b) XRD patterns and (c) the enlarged peaks from 25 to 45° of Figure 1b, d) Raman spectra of IrO₂/S-450 and IrO₂, e) Ir-L_{III}-edge XANES spectra of IrO₂/S-x, f) k³-weighted Fourier transforms of EXAFS spectra collected at the Ir-L_{III}-edge of the IrO₂/S-x, g) 3D wavelet transform maps of EXAFS signals of IrO₂/S-450 and IrO₂, h) Diagram of S-doping induced Ir-O coordination changes and (i) O K-edge XAS of IrO₂/S-450 and IrO₂.

through the reduced electrostatic attraction. In addition, the presence of Ir–S bonds (peak at 2.35 Å, $N = 0.8$) confirms S incorporation into the lattice. The reduced Ir–O coordination and the alteration of the Ir–O bond angle induced by S doping create a new energetic landscape for the oxygen species in the IrO₂ lattice (Figure 1h). As the Ir–O bonds are weakened and the lattice structure is distorted, some oxygen atoms exhibit different electronic and bonding environments compared to the regularly coordinated oxygen atoms, thus probably giving rise to the O_{NB} states. X-ray absorption spectroscopy characterization at the O K-edge was further conducted to verify the presence of O_{NB}. As shown in Figure 1i, 532 and 535 eV correspond to the hybridized states of the O 2p and the Ir 5d and 4 sp orbitals, respectively. Interestingly, the sample with S doping has an additional small peak at 528.9 eV, which is associated with holes created by O_{NB} states.^[6b,15] This peak is a direct consequence of the structural and electronic changes observed in the EXAFS analysis.

XPS was employed to comprehensively investigate the surface electronic structure and chemical states of the IrO₂/S-*x* (Tables S2–S4, Supporting Information), providing valuable insights that complement the structural information obtained from EXAFS. The Ir 4f XPS spectra of IrO₂/S-*x* samples (Figure S13a, Supporting Information) show distinct changes in the binding energy of Ir as a function of annealing temperature. For pure IrO₂, the Ir 4f doublet binding energies (Ir 4f_{7/2} and Ir 4f_{5/2}) appear at 62.7 and 65.7 eV, corresponding to the Ir⁴⁺ oxidation state in the rutile structure.^[16] In the S-dopant samples, as the annealing temperature increases from 300 to 450 °C, the binding energy of Ir initially decreases, which can be attributed to the electron-donating effect of S. This electron-rich environment effectively reduces the oxidation state of Ir, as reflected by the negative shift in the binding energy of the Ir 4f peaks. However, when the annealing temperature reaches 500 °C, the binding energy of Ir returns to a value close to that of pure IrO₂. The S 2p XPS spectrum (Figure S13b, Supporting Information) provides crucial information about the chemical environment of S in the IrO₂/S-*x* catalysts. The S 2p spectrum can be deconvoluted into two main peaks located at 163.8 and 168.6 eV, corresponding to Ir–S and S–O bonds,^[17] respectively. At 500 °C, the S signal disappears, probably indicating the decomposition of Ir–S and S–O into SO₂ and O₂.^[17a–d] The disappearance of S leads to the recovery of the Ir oxidation state, as shown in the Ir 4f XPS spectrum. This is consistent with the restoration of Ir oxidation state to pure IrO₂ levels in the XRD and XANES analysis.

To exclude the regulation effect of S–O bonds in the electronic structure of Ir active center, we synthesized a control catalyst containing only S–O species without Ir–S coordination (denoted as IrO₂/S–O) by reducing the thiourea ratio during synthesis (Details are provided in SI). The S 2p XPS spectrum of IrO₂/S–O (Figure S14a, Supporting Information) exclusively displays signals from S–O bonds, with no detectable Ir–S bonding features. Concurrently, the Ir 4f binding energy in IrO₂/S–O shows no deviation from that of the IrO₂/S–500 sample (Figure S14b, Supporting Information), indicating the unchanged electronic states of Ir in the absence of Ir–S coordination. XRD patterns (Figure S15a,b, Supporting Information) also reveal no lattice expansion in the IrO₂/S–O. The above results indicate that the electronic structure of Ir remains essentially unchanged in the presence of S–O bonds without Ir–S structures. This further confirms that

Ir–S coordination plays a dominant role in modulating the electronic structure of Ir.

2.2. Effect of S Doping on OER Performance

To explore the effect of S doping on the OER performance, linear sweep voltammetry (LSV) measurements were carried out on the IrO₂/S-*x* electrocatalysts in 0.5 M H₂SO₄ solution. Among these catalysts, the IrO₂/S-450 exhibits the highest OER activity with an overpotential (η_{10}) of only 276 mV at a current density of 10 mA cm^{−2}. Such an overpotential is ≈86 mV lower than that of our IrO₂ control sample (Figure 2a), indicating a significant enhancement in OER activity. Figure S16a,b (Supporting Information) presents the mass-normalized polarization curves and mass activity at 1.53 V vs. RHE for the IrO₂/S-*x* catalysts, respectively. The mass activity of IrO₂/S-*x* follows the same trend as the apparent activity, and it exhibits a volcano-type variation with increased heat treatment temperature. To be specific, the IrO₂/S-450 achieves optimal OER performance via the electron-donating effect of S (moderate Ir–S interaction), while excessive heat treatment in the IrO₂/S-500 leads to complete dissociation of Ir–S bonds (via S oxidation), and hence to the declined activity. For IrO₂/S-300 and IrO₂/S-350, S 2p XPS shows the abundant Ir–S complex, confirming the insufficient oxidation of Ir precursor, thereby exhibiting poor OER activity. To further assess the intrinsic activity, the turnover frequency (TOF) @ 1.53 V (vs. RHE) and electrochemical specific area (ECSA)-normalized current densities (j_{ECSA}) of the IrO₂/S-*x* electrocatalysts were calculated, and the relevant data are presented in Figures S17 and S18 (Supporting Information). As illustrated in Figure 2b, both TOF and j_{ECSA} also exhibit a volcano-type trend with the heat treatment temperature. The IrO₂/S-450 sample reaches the volcanic apex, displaying the highest intrinsic activity among all the tested samples. Cyclic voltammetric (CV) testing was performed to investigate the dynamic evolution process of the Ir sites. Figure S19a (Supporting Information) depicts the stable CVs of the IrO₂/S-450 and IrO₂ in 0.5 M H₂SO₄ in the potential range of 0 to 1.5 V (vs. RHE). A well-defined redox peak corresponding to the Ir⁴⁺/Ir⁵⁺ couple is observed for the IrO₂/S-450 at 1.194 V, which is absent in the CV on the IrO₂ sample. Significantly, the Ir⁵⁺ species remains stable even after 2000 cycles (Figure S19b, Supporting Information), indicating that the incorporation of S promotes the formation of the highly active Ir⁵⁺ species.

In situ electrochemical impedance spectra (EIS) were applied to gain insight into the reaction kinetics at the electrode and electrolyte interface (Figure 2c; Figure S20, Supporting Information). Specially, the electron transfer response in the inner layer of the catalyst occurs at the high-frequency range of 10²–10⁴ Hz, whereas the charge transfer response at the catalyst-electrolyte interface is observed at the low-frequency range of 10^{−2}–10² Hz.^[18] Compared to IrO₂, the IrO₂/S-450 exhibits a lower phase angle in both high and low-frequency ranges, with these peaks decreasing more rapidly as the potential increases, which suggests an accelerated charge transfer on the IrO₂/S-450 interface. In situ Raman spectroscopy reveals distinct structural dynamics of Ir active sites in IrO₂/S-450 during the OER. As shown in Figure 2d and Figure S21 (Supporting Information), the characteristic 550 cm^{−1} peak (denoted δ , assigned to Ir–O stretching

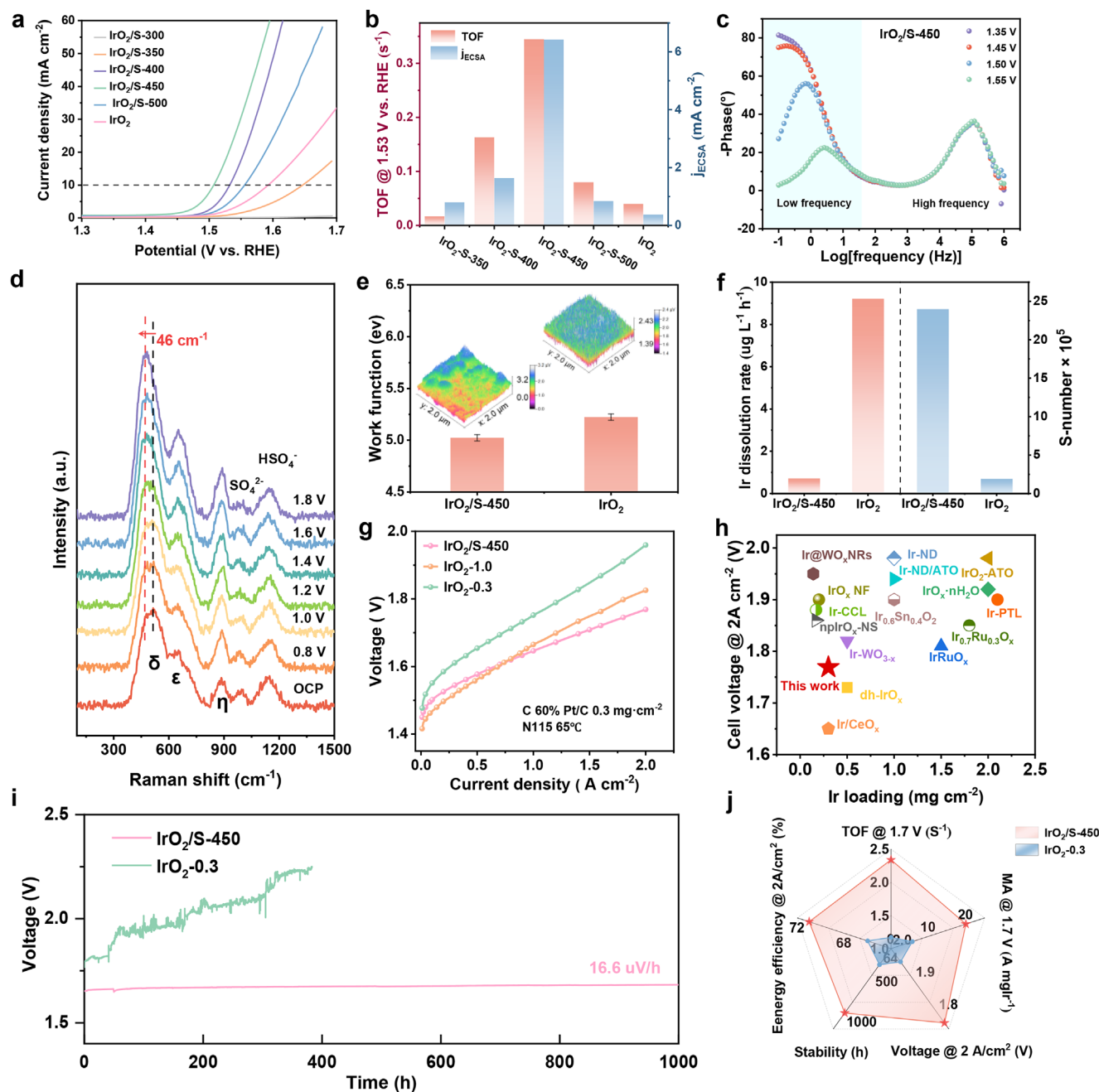


Figure 2. Effect of S doping on the electrochemistry of IrO₂/S. a) LSVs of the IrO₂/S catalysts in 0.5 M H₂SO₄ at a scan rate of 5 mV s⁻¹, b) Variation of TOF (@1.53 V/RHE) and j_{ECSA}, c) Bode phase plots of IrO₂/S-450, d) In situ Raman spectra at different potentials in 0.5 M H₂SO₄ of IrO₂/S-450, e) Comparison of work functions of different catalysts inferred from the contact potential difference values, f) ICP-MS analysis of the Ir dissolution rate for IrO₂/S-450 and IrO₂ after chronopotentiometric test, and the corresponding S-number, g) Polarization curves of the PEMWE with different anode catalysts at 65 °C, h) Comparison of PEMWE performance for various reported electrocatalysts, i) Stability test of PEMWE with IrO₂/S-450 at 1 A cm⁻² and at 65 °C and j) Comparison of MEA performance parameters of the IrO₂/S-450 and IrO₂-0.3.

vibrations) exhibits a significant redshift in IrO₂/S-450 compared to pristine IrO₂ as the applied potential increases. This redshift reflects accelerated oxidation of Ir sites in IrO₂/S-450,^[10b] further proving the fast electron transfer during the OER, which is also in good line with the in situ EIS results. To investigate the origin of the accelerated electron transfer, the surface potential was measured using Kelvin probe force microscopy (KPFM), as shown

in the inset of Figure 2e. The work functions inferred from the experimental contact potential difference values are 5.03 eV for IrO₂/S-450 and 5.19 eV for IrO₂ (Figure 2e, detailed calculation process can be seen in the SI).^[19] The relatively low work function of the IrO₂/S-450 suggests faster electron transport and thus enhanced OER kinetics. These results are further supported by the Tafel analysis presented in Figure S22 (Supporting Information).

The OER Tafel slope on the IrO₂/S-450 is 46.52 mV/dec, which is significantly lower than those of the other samples, indicating an enhanced OER kinetics, unambiguously corroborating the promotional effect of S doping on the OER activity.^[20]

Furthermore, the IrO₂/S-450 also exhibits excellent durability over 360 h test at a current density of 10 mA cm⁻² (Figure S23, Supporting Information). Inductively coupled plasma atomic emission spectroscopy (ICP-OES, Figure 2f; Table S6, Supporting Information) shows that the Ir dissolution rate of IrO₂/S-450 is only 0.72 µg L⁻¹ h⁻¹, which is much lower than that of our IrO₂ control sample (9.22 µg L⁻¹ h⁻¹). In addition, the stability-number (S-number) of the IrO₂/S-450 (Figure 2f, 2.4×10^6) is ca. 13 times higher than that of IrO₂ (1.9×10^5), further implying the superior structural stability of the IrO₂/S-450. Moreover, the binding energy of Ir (Figure S24, Supporting Information) for the IrO₂/S-450 after the durability test shifts to high values, indicating the partial oxidation of Ir sites, which is consistent with the CV and in situ Raman results. To explore the change of S atom after the durability test, we conducted stability tests on ionomer-free IrO₂/S-450 electrode in 0.1 M HClO₄ (Figure S25, Supporting Information) to eliminate the potential interference of SO₄²⁻ and SO₃H⁻. As illustrated in Figure S26 (Supporting Information), the Ir–S bond at 163.8 eV remains intact after the durability test, underscoring the structural robustness of the Ir–S coordination structure even under prolonged electrolysis conditions. Quantitative analysis of S–O and Ir–S components revealed that the Ir–S atomic percentage within the total S species (Ir–S + S–O) is ca. 20% for the initial IrO₂/S-450 catalyst and slightly increases to 25% for the post-OER sample. Meanwhile, the total S atomic content is 1.26% for an initial sample and 1.03% for a post-OER sample. Therefore, the calculated Ir–S contents are 0.252% for an initial sample and 0.257% for a post-OER sample, demonstrating that the S in Ir–S bonds remained nearly unchanged. This confirms negligible leaching of Ir–S species, with S loss primarily attributed to the dissolution of S–O components.

To validate the protective effect of O_{NB} on the Ir–O bond order, we performed XRD and Raman spectroscopy on IrO₂/S-450 before and after the OER. As shown in Figure S28a,b (Supporting Information), the Ir–O lattice structure remained essentially unchanged after the OER process, indicating the preservation of the crystal framework. The XAENS spectra of Ir and corresponding fitting results (Figure S29 and Table S8, Supporting Information) reveal no significant alteration in Ir–O coordination before and after the OER. This confirms that O_{NB} induced by S-doping plays a critical role in maintaining Ir–O bond order and ensuring structural stability during the OER.

To explore the application potential of the IrO₂/S-450 in the PEMWE device, the membrane electrode assembly (MEA) integrated with the optimized IrO₂/S-450 as the anodic catalyst with an Ir loading of only 0.3 mg_{Ir} cm⁻² is fabricated (Figure S30, Supporting Information). As demonstrated in Figure 2g, the MEA featuring IrO₂/S-450 and Nafion 115 membrane achieves a cell voltage of 1.769 V at 2.0 A cm⁻², which is 57 and 190 mV lower than those of MEAs with IrO₂-1.0 and IrO₂-0.3, respectively. Noteworthy, when using a thinner membrane (e.g., Gore, 80 µm), the cell voltage can further decrease to 1.690 V at 2.0 A cm⁻² (Figure S31, Supporting Information). The mass activity on the IrO₂/S-450 within the MEA (Nafion 115) at a cell voltage of 1.7 V is 20.0 A mg_{Ir}⁻¹, which is ca. 4.2 and 3.5 times higher than that on IrO₂-

1.0 (4.7 A mg_{Ir}⁻¹) and IrO₂-0.3 (5.7 A mg_{Ir}⁻¹) respectively, again attesting the boosted OER activity on the IrO₂/S-450 (Figure S32, Supporting Information). This performance also surpasses the most of reported Ir-based catalysts (Figure 2h; Table S9, Supporting Information) even at an ultra-low Ir loading of 0.3 mg cm⁻² (SI).^[21] Moreover, the energy efficiencies of the PEM electrolyzers using IrO₂/S-450 catalyst reach 76, 74, and 71% at 1.0, 1.5, and 2 A cm⁻² respectively, outperforming those of IrO₂-1.0 and IrO₂-0.3 (Figure S33a and Table S10, Supporting Information) at the low heating value (LHV). As shown in Figure S33b (Supporting Information), even under the assumption that all metal atoms in the IrO₂/S-450 catalyst are active for OER, the lower bound estimate of the TOF at 1.7 V reaches 2.49 O₂ s⁻¹, which is 8.3 and 35.6 times higher than those of IrO₂-1.0 and IrO₂-0.3, respectively (Table S11, Supporting Information).

Since the long-term stability of the MEA is a prerequisite for practical application in PEMWE, the stability of the IrO₂/S-450-based MEA was evaluated at a constant current density of 1.0 A cm⁻². As presented in Figure 2i, the MEA with an Ir loading of 0.3 mg cm⁻² exhibits encouraging stability over 1000 h with a decay rate of ca. 16.6 µVh⁻¹, significantly superior to that of IrO₂-0.3. We analyzed the catalytic layer structure of the MEA after stability testing. As shown in Figure S34a–d (Supporting Information), the catalytic layer structure with IrO₂/S-450 remained unchanged after 1000 h stability testing. In contrast, the IrO₂ catalyst exhibited pronounced structural collapse in its catalytic layer (Figure S34e–h, Supporting Information). Additionally, we characterized the electronic structures of Ir and S in the MEAs before and after stability testing. As depicted in Figures S35a,b (Supporting Information), the Ir 4f binding energies of both two catalysts shifted positively after 1000 and 400 h of testing, respectively, indicating further oxidation of Ir to higher valence states. Notably, Ir–S coordination within the IrO₂/S-450 catalyst persisted after the reaction (Figure S35c, Supporting Information), confirming its structural robustness. These results highlight the potential of realizing practical applications of anode catalysts in PEMWE with low Ir usage, low energy consumption, and high stability (Figure 2j), all of which are largely attributed to the beneficial effects of S doping on the OER activity and stability of the IrO₂ catalyst.

2.3. Mechanistic Insights into S-Doping Induced Pathway Switching

To further investigate the proton-electron transfer processes occurring at the electrodes, the pH-dependent activities of the IrO₂/S-450 and IrO₂ catalysts were initially measured. LSV measurements at different pH (pH 0, 0.25, 0.5, 0.75, and 1) for the IrO₂/S-450 and IrO₂ are shown in Figure S36 (Supporting Information). Moreover, the proton reaction order ρ^{RHE} derived from the partial derivation of current density (in log scale) at a specific voltage (1.53 V vs. RHE) with respect to pH values is given in Figure 3a. Clearly, the IrO₂/S-450 exhibits a heightened sensitivity to the proton concentration, as indicated by a high ρ^{RHE} value of 0.64. This observation implies a pH-dependent behavior of the OER kinetics, thereby indicating a LOM-dominated mechanism for the IrO₂/S-450.^[22] Conversely, ρ^{RHE} of IrO₂ drops to 0.18, suggesting the involvement of the AEM pathway.^[22] OER

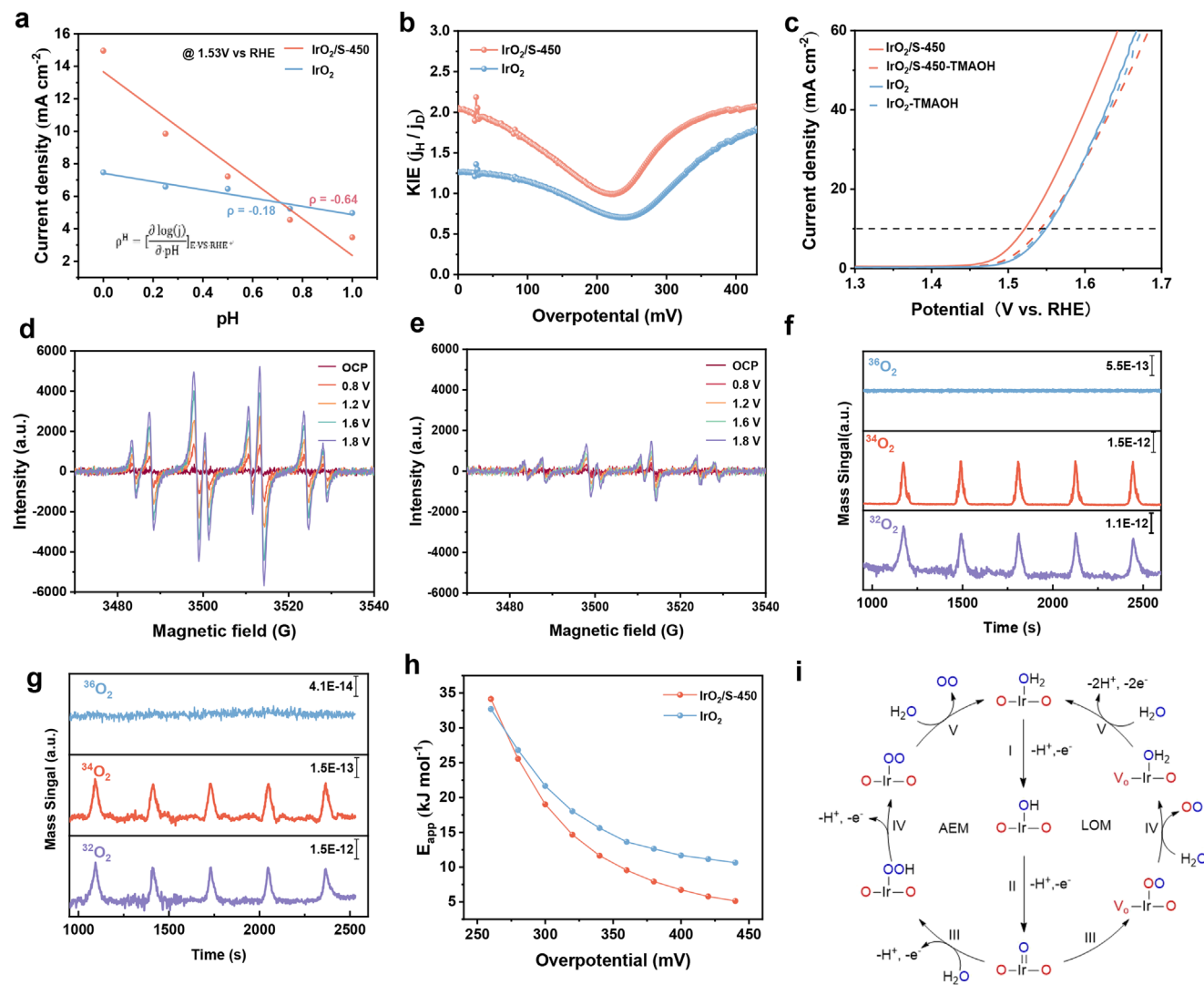


Figure 3. Exploration of OER pathway. a) OER current density at 1.53 VRHE plotted in log scale as a function of pH, from which the proton reaction orders ($\rho = (\partial \log(j)/\partial \text{pH})_E$) were calculated, b) KIE values at different overpotentials, c) LSVs of the IrO₂/S-450 and IrO₂ at different solutions, DMPO spins trapping EPR in O₂^{•-} of (d) IrO₂/S-450 and (e) IrO₂, DEMS signals of O₂ products for the ¹⁸O-labeled (f) IrO₂/S-450 and (g) IrO₂ in 0.5 M H₂SO₄ in H₂¹⁶O, h) The corresponding E_{app} data are calculated from LSVs at different temperatures and (i) Schematic illustration of AEM and LOM on the IrO₂/S-450.

activities were also tested in proton (0.5M H₂SO₄ in H₂O) and deuterium (0.5M D₂SO₄ in D₂O) electrolytes to investigate the kinetic isotope effect (KIE) (Figure S37, Supporting Information). KIE values were obtained based on the ratios of OER current density for hydrogen to that for deuterium at the same overpotential (Figure 3b). The IrO₂/S-450 exhibits an obviously larger KIE value compared with IrO₂, indicating a stronger proton kinetic dependency.^[23] The involvement of lattice oxygen intermediates was probed using tetramethylammonium hydroxide (TMAOH), which selectively binds negatively charged *O¹⁻ intermediates in LOM.^[24] As shown in Figure 3c and Figure S38 (Supporting Information), the OER activity of the IrO₂/S-450 is significantly reduced after adding the TMAOH. This inhibition confirms the presence of *O¹⁻ intermediates in the IrO₂/S-450, a hallmark of LOM. The lack of response in IrO₂ suggests the dominance of *OOH intermediates in AEM. To further detect electrochemi-

cal reaction intermediates, in situ EPR measurements were conducted. As shown in Figure 3d, no radical signals were detected at open-circuit potential. With the increase of applied potential, strong signals of 5,5-dimethyl-1-pyrroline N-oxide (DMPO), attributed to DMPO-O₂^{•-} adducts, were observed.^[25] This indicates that O₂^{•-} is generated in situ during the OER process on the IrO₂/S-450. These radicals serve as products of lattice oxygen transitioning from a low oxidation state (O²⁻) to a higher oxidation state. In contrast, the signal intensity of the O₂^{•-} radical in IrO₂ is extremely weak (Figure 3e), indicating that only a negligible amount of lattice oxygen is oxidized and that the AEM pathway predominates on IrO₂. Direct evidence for lattice oxygen participation was obtained via in situ ¹⁸O-labeling DEMS (Figure 3f). The IrO₂/S-450 and IrO₂ were potentiostatically pretreated in 0.5 M H₂SO₄ electrolytes with H₂¹⁸O as the solvent (details are shown in the Supporting Information and Figure S39,

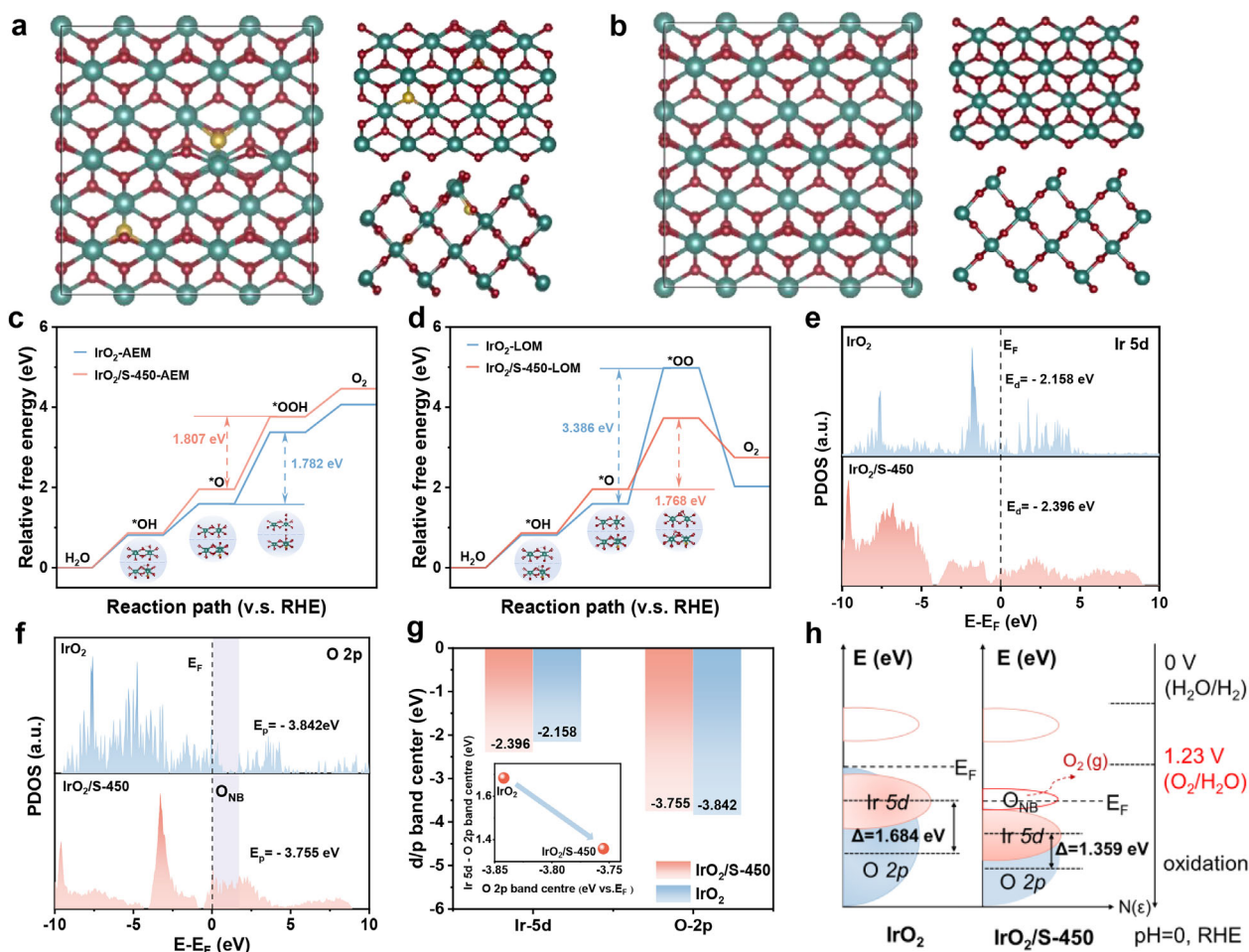


Figure 4. DFT calculations. Atomic structure of the (a) IrO₂/S-450 and (b) IrO₂ surface, the Ir, O, and S atoms are shown in blue, red, and yellow, respectively. (c) Gibbs free energy illustration by IrO₂/S-450 and IrO₂ during the OER by the AEM pathway and (d) by the LOM pathway (e) PDOS plots of Ir 5d states in IrO₂/S-450 and IrO₂ (f) PDOS plots of O 2p states in IrO₂/S-450 and IrO₂ (g) Difference between the Ir 5d-band center and O 2p-band center versus the O 2p band center relative to the Fermi level for IrO₂/S-450 and IrO₂ (h) Schematic rigid band diagrams of IrO₂ and IrO₂/S-450. The position of the O₂/H₂O redox couple at pH 0 (2H₂O → O₂ + 4H⁺ + 4e⁻) is 1.23 VRHE.

Supporting Information). For the IrO₂/S-450, ¹⁶O¹⁸O (*m/z* = 34) accounts for 38% of O₂ products, indicative of the lattice oxygen contribution. The absence of ¹⁸O¹⁸O (*m/z* = 36) excludes the possibility of adjacent lattice oxygen coupling, confirming a single-site LOM mechanism.^[16a] In contrast, ³²O₂, the gas product for the IrO₂ contains 0.38% ³⁴O₂ (Figure 3g). Therefore, the detected amount of ¹⁸O in OER products is 0.19%, which can be attributable to the natural isotopic abundance (≈0.2%) of ¹⁸O in water. The result suggests that IrO₂ strictly follows a traditional AEM without the involvement of lattice oxygen in the catalytic cycle. These results unambiguously demonstrate that S doping switches the OER pathway from AEM to LOM on IrO₂. To characterize the binding ability of Ir sites toward OH species for the two catalysts, we conducted methanol oxidation experiments. The difference in current density between the methanol oxidation reaction (MOR) and OER is proportional to the number of transferred charges, and this difference can be quantified by calculating the filled area between the two curves.^[26] Interestingly, as shown in Figure S40 (Supporting Information), compared to the IrO₂ catalyst, the polarization curves of IrO₂/S-450 before and

after methanol addition are not much different, indicating the weak OH adsorption capacity on the surface of IrO₂/S-450. This weak adsorption facilitates the desorption of OH during the LOM pathway.

We further conducted a temperature-dependent kinetic analysis of both IrO₂/S-450 and IrO₂ catalysts to extract the apparent activation energy (*E*_{app}) of OER. The OER polarization curves for both catalysts were recorded in 0.5M H₂SO₄ solution over a temperature range of 25–65 °C (Figure S41, Supporting Information). The *E*_{app} values at specific overpotentials were calculated from the slope of the Arrhenius plots (Figure 3h; Figure S42, Supporting Information).^[27] Notably, at low overpotentials (*η* < 280 mV), IrO₂ exhibits lower apparent activation energy (*E*_{app}) due to facile *OOH formation in AEM. However, as the overpotential increased, the *E*_{app} of IrO₂/S-450 decreased to a significantly lower value than that of IrO₂. This crossover suggests that S doping lowers the energy barrier for lattice oxygen activation at high *η*, aligning with the DFT-predicted (Figure 4d). Collectively, the results presented herein elucidate that S doping triggers an OER pathway switch from the AEM to LOM. To mechanistically

rationalize this observation, we propose a detailed reaction pathway involving five elementary steps—comprising four electrochemical steps and one non-electrochemical O_2 desorption step—for both AEM and LOM mechanisms, as schematically illustrated in Figure 3i.

2.4. Theoretical Investigation of the Mechanism Switch and Stabilization by S Modulation

To unravel the origin of the observed performance enhancement, density functional theory (DFT) simulations were performed on undoped and S-doped IrO_2 (Figure 4a,b). The Gibbs free energy profiles (Figure 4c,d) reveal a striking contrast between AEM and LOM pathways. For AEM, $IrO_2/S-450$ exhibits a slightly higher ΔG of 1.807 eV compared to IrO_2 (1.782 eV), indicating suppressed $*OOH$ formation. However, in the LOM pathway, $IrO_2/S-450$ achieves a significantly lower ΔG of 1.618 eV, representing a 1.768 eV reduction relative to IrO_2 (3.386 eV). Meanwhile, this energy barrier is also lower than that of the reaction in the AEM pathway (1.807 eV). This finding directly corroborates the experimental observation that the LOM pathway is kinetically favored on the S-doped catalyst surface, whereas the AEM pathway remains dominant on undoped IrO_2 . To unravel the electronic origin of this mechanistic shift, projected density of states (PDOS) analyses were conducted to examine the orbital interactions between Ir 5d and O 2p states. S doping shifts the Ir d-band center from -2.16 eV (IrO_2) to -2.29 eV ($IrO_2/S-450$) (Figure 4e), weakening $*OOH$ adsorption and suppressing AEM. Experimental validation was provided by XPS valence band spectra (VBS, Figure S43, Supporting Information), which revealed a negative shift in the d-band center from -2.16 eV (IrO_2) to -2.29 eV ($IrO_2/S-450$), consistent with DFT prediction. Analysis of the O 2p PDOS (Figure 4f) showed a notable upward shift in the O 2p band center from -3.842 eV (IrO_2) to -3.755 eV ($IrO_2/S-450$), reducing the energy barrier for lattice oxygen activation. Besides, the energy gap between E_d and E_p is reduced from 1.684 to 1.359 eV upon S doping, indicating enhanced Ir–O covalent bonding (insert of Figure 4g). This endows the antibonding states below the Fermi level with more O character. Additionally, analysis of the O 2p PDOS reveals a substantial population of empty O_{NB} states above the Fermi level (E_F) in $IrO_2/S-450$ (Figure 4f).^[6a,28] These states act as ESA, diverting charge transfer from direct Ir–O bond cleavage to high-energy O_{NB} states (Figure 4h). By decoupling charge transfer from Ir–O bond breaking, the integrity of the metal-oxide framework is preserved during the LOM pathway, thereby maintaining the catalyst's structural stability. This mechanism represents a key structural advantage conferred by S doping, rationalizing the exceptional durability observed in $IrO_2/S-450$ under prolonged OER operation.

3. Conclusion

In summary, we present an S-mediated IrO_2 OER catalyst that exhibits exceptional performance and stability in PEMWEs, with the device performance surpassing that of the 2026 US-DOE target. The most important finding is that the doping of S not only can switch the OER pathway on IrO_2 from the conventional AEM

to LOM, thereby breaking the linear relationship and increasing the intrinsic activity but also generates an O_{NB} band that preserves the Ir–O bond order, consequently improving the stability of the LOM pathway. As a result, the optimal $IrO_2/S-450$ catalyst achieves a cell voltage of 1.787 V at 2.0 A cm^{-2} and operates steadily at 1.0 A cm^{-2} for 1000 h with a low Ir usage. Collectively, this study establishes a rational design strategy for low-Ir-usage OER catalysts capable of withstanding the harsh operational conditions of PEMWEs. These findings provide a blueprint for developing high-performance electrolysis technologies aligned with global decarbonization goals.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

Acknowledgements

This work was supported by the National Key Research and Development Project (2021YFB4000200), the Strategic Priority Research Program of the Chinese Academy of Sciences (XDA0400301), Shanghai Zhang Jiang S&T Innovation Center (ZJ2022-ZD-008), and Shanghai Rising-Star Program (24QA2710500, Shanghai Scientific and Technological Committee), the Talent Plan of Shanghai Branch of the Chinese Academy of Sciences (CASSHB-QNPD-2023-006). We would also like to acknowledge the BL14W1 and BL20U1 beamlines at the Shanghai Synchrotron Radiation Facility (SSRF). The authors extend their gratitude to Mr. Su Yibo and Huang Jing from Scientific Compass (www.shiyanjia.com) for providing invaluable assistance with the XAS analysis.

Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

iridium, lattice oxygen mechanism, oxygen evolution reaction, oxygen non-bonding state, PEM water electrolysis

Received: April 21, 2025
Revised: June 4, 2025
Published online: June 23, 2025

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